Mumbai Education Trust's

INSTITUTE OF ENGINEERING, NASHIK .

COMPUTER ENGINEERING DEPARTMENT

Subject : DS **ASSIGNMENT NO – 02** **Unit : II**

1. Write a short note on **Storage Representation of an array**.
2. What are advantages & disadvantages of **Sequential Organization** of data structure?
3. Derive **Address calculation formula for one-dimensional array** with one example.
4. Explain **Two-dimensional arrays** with row and column major implementation. Explain address calculation in both cases with example.(or Explain **row major & column major representation of Arrays** in computer memory)
5. What is **Sparse matrix** ? Explain with suitable example.? Write algorithm to perform addition of two matrix.
6. What is **difference between simple and fast transpose** of sparse matrix? Write an algorithm to find simple transpose of sparse matrix with time complexity.
7. Explain **Fast Transpose of sparse matrix** with suitable example, Discuss time complexity of fast transpose.
8. Explain **Linear search** and **binary search** with Algo. and example. State its time complexity and **Compare linear and binary search(time and space complexity)**

   OR Explain in brief any **three searching techniques**. What is the time complexity of these techniques?
9. Explain **Fibonacci, sentinel & Index sequential search with suitable example.**
10. Explain the **Algorithm of Quick sort** with suitable example. Discuss its time complexity and space complexity.
11. Write **short note on Stability of Sorting**. Compare bubble, insertion and selection sort with one example and discuss time complexity
12. Write pseudo '**Python' algorithm (recursive) for Binary search.** Apply your algorithm on the following numbers stored in array from A[0] to A[10]

    9, 17, 23,38,45,50,57,76,90,100 to search numbers 10 , 23 & 100

13. Write an **algorithm to search an element in array A using binary search** technique. Show stepwise search of Key 10 in 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11. Is there any pre-processing of data required before starting binary search or what is its precursor requirement?

14. Sort given array by using **bubble sort method**: 64, 34, 25, 12, 22, 11, 90. Show step by step execution for all the passes highlighting "swap" and "No Swap" situations. How many passes are required to sort an array of N elements using bubble sort?

15. Explain **insertion sort algorithm** and sort the given list using insertion sort :

List : 7, 4, 10, 6, 3, 12, 1, 8, 2, 15, 9, 5

List : 55, 85, 45, 11, 34, 05, 89, 99, 67

Discuss its time complexity and space complexity.

16. Sort given array by using **selection sort method** 50, 23, 03, 18, 9, 01, 70, 21, 20, 6, 40, 04. Show step by step execution of all passes. What is the best and worst case time complexity of selection sort?

17. Explain **quick sort and Sort** the following numbers using quick sort :

State its time complexity and space complexity.

List : 27, 76, 17, 9, 57, 90, 45, 100, 79.

List : 39, 09, 81, 45, 90, 27, 72, 18

List : 25, 82, 17, 23, 38, 7, 64, 86, 21

List : 15, 08, 20, -4, 16, 02, 01, 12, 21, -2

18. Explain **Merge sort**. Sort following example using Merge Sort :

List : 18, 13, 12, 22, 15, 24, 10, 16, 19, 14, 30.

List : 55, 85, 45, 11, 34, 05, 89, 99, 67

Discuss its time and space complexity.

********** **Best of Luck** **********

Study material provided by: Vishwajeet Londhe

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